

Sequential Oxygen Mismatch from Skeletal Muscle to Prefrontal Cortex Underpins the Rate of Exhaustion during All-Out Exercise

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ABSTRACT

HARTMAN, M. E., M. KANTOR, K. THORNHILL, S. L. REINER, B. J. WINN, M. KRAMER, R. W. PETTITT, and B. S. KIRBY. Sequential Oxygen Mismatch from Skeletal Muscle to Prefrontal Cortex Underpins the Rate of Exhaustion during All-Out Exercise. *Med. Sci. Sports Exerc.*, Vol. 57, No. 11, pp. 2368–2381, 2025. **Purpose:** We tested the overarching hypothesis that the expended rate of work above critical power (W' Balance) during all-out whole-body exercise is related to a decline in prefrontal cortex (PFC) oxygenation secondary to an organized systemic outstripping of muscle O_2 supply relative to O_2 demand. **Methods:** We concomitantly measured ($n = 16$ males) skeletal muscle O_2 saturation (vastus lateralis near-infrared spectroscopy (NIRS); $\%S_{mO_2}$), pulmonary O_2 uptake ($\dot{V}O_2$), and hemoglobin (Hb) differential ($\Delta[O_2Hb - HHb]$) as an index of PFC O_2 mismatch ($\%O_2$) via functional NIRS bilaterally in the ventrolateral (VLPFC), dorsolateral (DLPFC), and orbitofrontal (OFC) cortices during brief all-out cycling exercise (highest instantaneous power for 3 min). **Results:** All-out exercise evoked significant changes in $\%S_{mO_2}$ ($\Delta -28.8\% \pm 14.1\%$), $\dot{V}O_2$ ($\Delta 27.7\% \pm 10.3\%$), and global $\%O_2$ ($\Delta -7.6\% \pm 4.8\%$). Decreases in regional $\%O_2$ were greater in the VLPFC ($\Delta -10.9 \pm 6.1 \mu M$) versus DLPFC ($\Delta -4.8 \pm 4.5 \mu M$) or OFC ($\Delta -5.9 \pm 4.2 \mu M$). Spatiotemporal analysis by O_2 measurement location revealed a steep rate of change transition phase followed by a maximal sustaining plateau, and progression of this pattern occurred sequentially first in muscle (~ 13 s) $\rightarrow$ pulmonary (~ 44 s) $\rightarrow$ PFC (~ 80 s). Transition phase O_2 indices were strongly correlated with the rate of W' Balance expended (muscle, $R^2 = 0.91$; pulmonary, $R^2 = 0.997$; PFC, $R^2 = 0.968$), with crossover between regional O_2 mismatches occurring at the same $\%W'$ Balance (end muscle = 71% vs start pulmonary = 65%, $P = 0.56$; end pulmonary = 26% vs start PFC = 30%, $P = 0.83$) and end PFC transition phase occurring at complete depletion of W' (end PFC = -0.9%). **Conclusions:** We conclude that whole-body all-out exercise tolerance may arise from a progressive O_2 mismatch from skeletal muscle to the brain. **Key Words:** EXERCISE TOLERANCE; HIGH INTENSITY, CRITICAL POWER, W'

INTRODUCTION

In bioenergetic terms, the quantification of exertional fatigue leading to exhaustion has been characterized by the hyperbolic model of the power–duration relationship (1–3), which has made substantial progress in predicting athletic performance

(4,5). In this model, critical power (CP) is the external work rate that coincides with the physiological measure of the maximal metabolic steady state that delineates the domains of heavy- and severe-intensity exercise. The work that can be performed in the severe-intensity domain or above CP is mathematically encapsulated by the secondary parameter derived from the hyperbolic model, a high-intensity work capacity index, termed W' -prime (W''). The rate of W' expenditure during

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exercise above CP is predictable, and modeling the balance of $\dot{W}$ remaining ($\dot{W}$ Balance) across intensity domains acts to reflect the *net* outstanding amount of high-intensity work capacity as well as accurately forecasts the dynamic fatigue process and time to exhaustion for a given severe-intensity exercise (1,6). Despite the model's predictive accuracy, a comprehensive understanding of the mechanistic basis of $\dot{W}$ remains partial, and research has predominantly focused on peripheral factors (e.g., high-energy muscle phosphates, muscle glycogen, myoglobin, the $\dot{V}O_2$ slow component, and type II muscle fiber recruitment) versus central factors (e.g., central nervous system related) (2,3,7–9). Consequently, the relation of $\dot{W}$ expenditure to central neurophysiological control remains largely unclear.

Tissue O_2 availability appears broadly linked to $\dot{W}$ and the sustainable rate of severe-intensity exercise (6,8,10). Local muscle blood flow and muscle O_2 saturation become insufficient during exercise above CP (10,11). Moreover, we recently observed during whole body exercise to exhaustion that skeletal muscle O_2 saturation progressively and predictably declines within both active and non-active muscle and is closely related to $\dot{W}$ (or D' during running) (10). Indeed, severe-intensity exercise eliciting the upper boundary of achievable pulmonary O_2 uptake and peak muscle fiber recruitment (7) is well understood to couple with a hemodynamic juxtaposition between maintenance of blood pressure and sufficient oxygen delivery to active muscle (12). In this paradigm, exorbitant peripheral vasodilation challenges the pumping capacity of the heart such that baroreflex-mediated sympathetic activation restrains tissue blood flow and coincides with a decrease in oxygenation of muscle and the cerebral cortex (1,7,10,12). This may importantly suggest that overall systemic control of hemodynamics and oxygen delivery is paramount, either directly or indirectly, in controlling the rate of $\dot{W}$ Balance expended and maximal intensity work capacity.

Recent reports highlight a potentially dynamic interplay between skeletal muscle and brain oxygenation (specifically the cerebral cortex), suggesting that both could be compromised at the point of exhaustion (13). Accordingly, work rates performed above CP generally appear to show a decline in global PFC oxygenation; however, a majority of these studies have examined exercise in an ascending ramp (14–16) or a constant-paced intensity to exhaustion (12,17), with no published reports to date examining a relation to the rate $\dot{W}$ Balance expended or all-out patterned exercise. In this regard, and in contrast to all-out exercise (7), maximal power output and/or peak motor unit recruitment would not occur until the final segments of an ascending ramp or a constant-paced exercise; thus, it remains unclear whether it is simply execution of work rates above CP *per se*, or secondary homeostatic perturbations consequent to the onset of such exercise that may truly govern PFC involvement in the maximal limits of exercise tolerance. All-out exercise (1,7,18), or effectively, a prolonged (energetically) supramaximal sprint effort that results in a rapid peak power output but ultimately descending intensity progression, is a very specific type of exercise in terms of reversing the profile of overall intensity, muscle activation, motivation, and

mental engagement. Therefore, such exercise may be one way to probe higher-level brain processing regions (such as the PFC) and the associated integration with local and systemic control of skeletal muscle O_2 availability.

Emerging evidence supports that exercise tolerance is modulated by both neural processes (14,19,20) and psychological states (21,22). The perception of effort and the sense of exertional fatigue are in part regulated by decisional and motivational processes within the PFC (19,20,23,24). Frontal lobe subregion-specific areas of the PFC, such as the ventrolateral PFC (VLPFC), the dorsolateral PFC (DLPFC), and the orbitofrontal cortex (OFC), govern several other deterministic psychological states that are essential for exercise performance, including pace regulation (23), motivation (25), alertness and attentional focus (26), goal-oriented decision-making (27), motor planning (28), task reappraisal (29), and emotional regulation (30). For example, a person's motivational state to sustain effort during exhaustive exercise is directly related to their ability to cognitively suppress the aversive sensations associated with the strain of the exercise (31), with the DLPFC and OFC having been previously linked to exhaustive exercise termination (20,32) and the VLPFC revealing changes in brain activity following an exercise intensity transition from moderate to high intensity (33). Moreover, severe-intensity exercise performed within a graded intensity ramp test suggests that both affect (i.e., emotional state) and PFC oxygenation are not only subregion- but also hemisphere-specific (24,34). Accordingly, subregion and hemisphere areas within the PFC are not only paramount in regulating our thoughts, decisions, emotions, and actions, but also appear intricately linked to the control of exercise work capacity. Therefore, it is collectively important to examine: 1) psychological and neural influences during maximal effort exercise in tandem with other physiological but nonpsychological mediating variables, and 2) the underlying subregions and hemispheres within the PFC that may link physiological responses to decisions of human behavior.

In line with that proposed by Amann and Dempsey (35), temporally it follows that during exhaustive all-out exercise, peripheral changes within locally active skeletal muscle (e.g., inorganic phosphate accumulation) may precede metabosensitive group III/IV muscle afferent activation. In sequence, this may then result in global CNS efferent-mediated control of systemic hemodynamics, wherein cerebral oxygenation becomes compromised enough, or muscle acidosis augmented enough, to surpass a given individual critical threshold and elicit CNS efferent-mediated restriction of motoneuronal output (36). We postulate that the expended rate of work above CP (i.e., $\dot{W}$ Balance) during all-out whole-body exercise is related to a decline in subregional and hemisphere-specific PFC oxygenation secondary to an organized systemic outstripping of muscle O_2 supply relative to O_2 demand. To examine this, we concomitantly measured, in time series, a peripheral index of skeletal muscle O_2 saturation, a systemic index of whole body O_2 consumption, and central indices of frontal lobe tissue oxygenation across the entire PFC including the bilateral examination of the VLPFC,

DLPFC, and OFC during brief all-out whole-body cycling exercise consisting of an immediate and continuous display of the highest power output possible for an entire duration of 3 min. The chosen exhaustive all-out exercise protocol is known to induce a rapid and “prolonged window” of supramaximal metabolic perturbation, making it a powerful means by which to observe underlying processes linked to the maximal limits of exercise tolerance.

METHODS

Participants and ethical approval. Seventeen healthy adults were recruited for the present study. Specifically, the inclusion criteria were as follows: 1) 18 yr or older, 2) a self-classified recreational or competitive cyclist, 3) willing to exercise at vigorous intensity, 4) regularly engage in aerobic exercise (i.e., at least 3 d·wk⁻¹ for the last 3 months with >150 min weekly of moderate intensity or >75 min of vigorous intensity), 5) no previous lower extremity injuries, and 6) symptom-free from COVID-19 for the past 14 d and no history of significant cardiovascular or neurological health conditions. Although female volunteers were not specifically excluded per study design criteria, participants were recruited through local cycling clubs by flyer and resulted in all male volunteers, presumably because this sample pool is often majorly composed of men. All participants were instructed to maintain their regular sleep schedules, dietary intake, and caffeine consumption and to avoid high-intensity exercise 24 h before the test. The university’s institutional review board approved the study, and the participants provided written informed consent in accordance with the Declaration of Helsinki, except for registration in a database, prior to the start of the study. An effect size of 1.048 was determined from changes in O₂Hb from Subudhi et al. (15), and computed a requirement of at least 10 participants when using a 5% error and 80% power (G*Power software). For context, an estimated mean difference of $\Delta 5 \pm 5 \mu\text{M}$ in chromophore signal from an intervention is an effect size of 1.0, which in turn computes 10 participants based on 5% error and 80% power for a difference of means statistical *t*-test.

All-out cycling exercise. Participants performed 3 min of exhaustive all-out exercise (3MT) expected to induce a “prolonged” supramaximal metabolic demand (i.e., mechanical power output with an energetic requirement greater than what can be achieved via 100% $\dot{V}\text{O}_2$ max) on an electronically controlled cycle ergometer set with a fixed load (Monark LC6, Vansbro, Sweden) (1,7,18,37). The 3MT protocol includes extreme and severe-intensity domain power output, which results in a stimulus for 1) rapid attainment and sustained plateau of maximal $\dot{V}\text{O}_2$, 2) inexorable and significant depletion of skeletal muscle high-energy phosphate compounds and metabolite accumulation, 3) peak recruitment and activation of nearly all task-specific muscle fibers from exercise onset, 4) very high O₂ cost of fatigued muscle fibers per unit of work done, 5) and pronounced peripheral vasodilation and skeletal muscle hyperemia, sympathetic activation, maximal cardiac output, and compromised maintenance of blood pressure

(3,7,11,12,18). Thus, the chosen exercise protocol both rapidly elicits and extends the viewable window of underlying physiology during a maximal effort exercise in humans, collectively making it one of the most effective ways to study the processes underlying the maximal limits of exercise tolerance.

The load for the 3MT was derived from a series of regression equations, including height, body mass, and physical activity rating (37). Immediately following a 5-min warm-up at 50 W, each participant was instructed to pedal at their maximal cadence and to put forth a continuous maximal all-out effort for the entire duration of 3 min. Other than cadence, no performance-related feedback or time of test duration was provided. Power data were filtered through a preset, two-pass Butterworth filter with a 5-Hz cutoff (38). The end power (last 30 s) of the 3MT was used to determine CP (1,18), and the work performed above CP was used to calculate *W'* expressed in kilojoules (38).

PFC oxygenation. PFC oxygenation indices were measured via continuous wave functional near-infrared spectroscopy (fNIRS) (Brite 24; Artinis Medical Systems, Zetten, the Netherlands). This fNIRS system uses two wavelengths of near-infrared light (760 and 850 nm) to measure the optical absorption characteristics of oxyhemoglobin (O₂Hb) and deoxyhemoglobin (HHb) to estimate changes in relative concentration (μM) and has been used to measure oxygenation during exercise (39–41). Within the instrument, optical density data were converted into relative changes in hemoglobin (Hb) chromophore concentrations by way of the modified Beer–Lambert law. All parameter values were calibrated to system baseline prior to study measurement, and $\Delta[\text{O}_2\text{Hb}]$ and $\Delta[\text{HHb}]$ were exported using the proprietary software for offline preprocessing (OxySoft; Artinis Medical Systems). Thus, relative changes from a pre-exercise baseline of zero were determined for oxyhemoglobin (O₂Hb), HHb, total hemoglobin (O₂Hb + HHb; THb), Hb O₂ differential (O₂Hb – HHb; Hb Diff). Total Hb presents as an index of local tissue blood volume or flow. Hb Diff allows the identification of and acts as an index of PFC O₂ mismatch ($p_{\text{fc}}\text{O}_2$) or the relation of O₂ supply relative to O₂ demand within the specified measured brain region, where a decreasing (Hb Diff) signal reflects *net* cerebral PFC O₂ demand > O₂ supply, and an increasing signal reflects *net* cerebral PFC O₂ supply > O₂ demand (42–44). Per general near-infrared spectroscopy (NIRS) guidance (45), all chromophore data are provided; however, Hb Diff as $p_{\text{fc}}\text{O}_2$ is predominantly presented as the primary PFC interest variable throughout this report.

A total of 24 channels with 10 emitters and eight detectors were evaluated using the systems provided proprietary headcap and optode template. Participants were each fitted with a small, medium, or large headcap based on their head circumference. The proprietary Artinis headcap was placed on the skull by using the nasion, vertex (Cz), and inion as landmarks. The bi-hemisphere layout of the 24 channels covered the Broadman areas (BA) 10 and BA 11 corresponding to the OFC, BA 44, BA 45, and BA 47 corresponding to the VLPFC, and BA 9, BA 46, and the lateral segment of BA8 corresponding to the DLPFC. The channels were positioned to cover over these

regions following the modified combinatorial nomenclature of the 10–20 electrode positioning system (46), with locations measured using a Gulick tape. Specifically, the OFC included Fp2 and Fp1; the VLPFC included AF7, F7, AF8, and F8; and the DLPFC included F3, F4, AF3, AF4, and regions between F1 to FC3, and F2 to FC4 (47) (Fig. 1). All inter-optode distances were 30.0 mm. Age-specific differential pathlength factors ($4.99 + 0.067(\text{age}^{0.814})$) were calculated using the modified Beer–Lambert equation (48). The signal quality (i.e., light intensity) was examined prior to data recording (49). During pre-exercise setup, the Oxysoft software program makes a binary determination for out-of-range signals (i.e., yes/no). We corrected any out-of-range signals, which corresponded to specific optodes, by methods such as cleaning the hair beneath the optode(s) and increasing the pressure of the optode(s) on the scalp by tightening the headcap strap. This troubleshooting method follows what was recommended by the Oxysoft user manual and minimizes out-of-range signals. The software default data sampling rate was 25 Hz, and measurement was continuously recorded throughout the entire duration of the exercise. These procedures have demonstrated adequate reliability and validity during exercise (50,51).

Signal preprocessing was performed using the procedures recommended by Pinti et al. (52) and Sappia et al. (49) (MATLAB; The MathWorks, Natick, MA). Specifically, large movement-related artifacts (i.e., atypical signal inflections) were eliminated, and baseline shifts (i.e., instantaneous signal shifts) were corrected using the NIRS Analysis Package (53). The data segments that the movement-related artifacts affected were reconstructed using a piecewise low-order polynomial interpolation. A zero-lag finite impulse response bandpass filter was used to remove systemic physiological oscillations specific to exercise (i.e., 0.09–0.01 Hz) (52). A denoising algorithm removed any additional noise (e.g., amplitude spikes) unrelated to the underlying oxygenation signal (54). The data were converted into second-by-second values, reported as the median value of each 1-s epoch. Next, signal quality was checked by calculating the signal quality index (49), and poor-quality

signals below a threshold of 1.478 were removed. Multiple channels were collected and averaged into their corresponding brain regions; thus, excluding a poor-quality channel did not result in missing data for any brain regions. Outlier values three times the absolute deviation around the channel's median values were eliminated from further analysis (55). Poor-quality signals occurred within 16% of the channels.

Pulmonary oxygen uptake. Pulmonary O_2 uptake ($\dot{V}\text{O}_2$; $\text{mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$) was measured as an index of the rate of whole-body O_2 transport and utilization, where an increasing signal reflects an increase in metabolic O_2 demand as a net response for all metabolically active and engaged skeletal muscle. Pulmonary O_2 uptake was measured using a portable gas analyzer ($\dot{V}\text{O}_2$ Master; $\dot{V}\text{O}_2$ Master Health Sensors Inc., Vernon, BC, Canada) (56), which was calibrated before each test according to the manufacturer's guidelines. The analyzer and the high ventilation flow passage adaptor were connected to an over-nose mask (Hans Rudolph, Lenexa, KS), with lines feeding into a gas analyzer, obtaining breath-by-breath measurements. Breath-by-breath O_2 uptake was interpolated into 1-s intervals, and a 15-s moving average was applied to the data.

Skeletal muscle oxygen saturation. Skeletal muscle O_2 saturation ($\%\text{S}_{\text{mO}_2}$) was measured as an index indicating the balance of O_2 supply to O_2 demand within the measured skeletal muscle region, where a decreasing signal reflects *net* tissue O_2 demand > O_2 supply and an increasing signal reflects *net* tissue O_2 supply > O_2 demand. Accordingly, NIRS monitoring over skeletal muscle detects changes in the oxygenation of small blood vessels, capillaries, and intracellular sites of O_2 utilization and transport (e.g., myoglobin) (45). Right vastus lateralis skeletal muscle O_2 saturation was continuously measured using portable, wireless NIRS sensors (Moxy Monitor; Fortiori Design LLC, Hutchinson, MN; NIRS-Moxy). The unit relies upon four wavelengths of near-infrared light (680, 720, 760, and 800 nm) with inter-optode distances of 12.5 and 25.0 mm to assess the optical absorption characteristics of oxygenated Hb and myoglobin. NIRS-Moxy determines the ratio of the oxyhemoglobin concentration to the total Hb

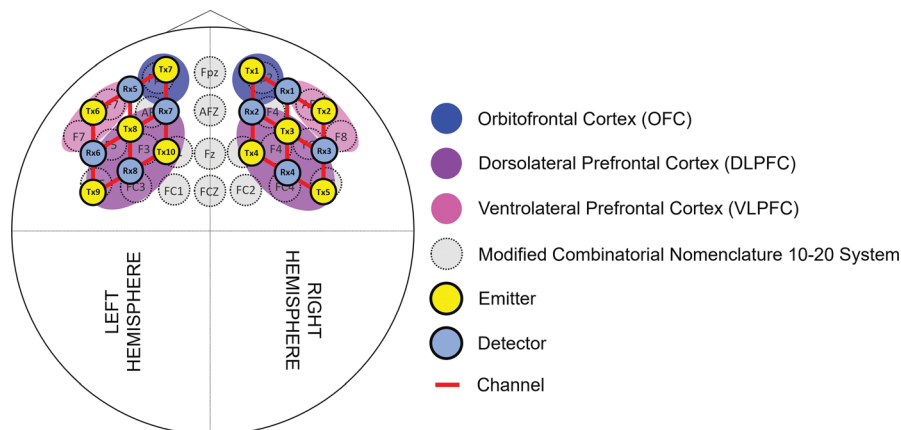


FIGURE 1—Schematic illustration for PFC fNIRS measurements. Headcap configuration for fNIRS corresponding to 18 optodes (10 emitters, yellow circles; 8 detectors; light blue circles) and the 24 channels (pairings, red line) is presented with an overlay of the bilateral regions of the OFC (blue region with two channels in each hemisphere), DLPFC (purple region with six channels in each hemisphere), and VLPFC (pink region with two channels in each hemisphere). Regions were defined by the Modified Combinatorial Nomenclature 10–20 system (gray circles).

concentration in the muscle and outputs this value as a percentage via the following equation:

$$\%S_{mO_2} = \left(\frac{\text{oxygenated hemoglobin} + \text{myoglobin}}{\text{total amount of hemoglobin} + \text{myoglobin}} \right) \times 100$$

where m stands for muscle, and % is expressed on an *a priori* 0% to 100% scale. Thus, $\%S_{mO_2}$ is computed similarly to tissue O_2 saturation (S_{tO_2}) provided by other NIRS manufacturers, aside from the increased specificity to skeletal muscle for NIRS-Moxy (45). The NIRS-Moxy was positioned over the right vastus lateralis 15 cm proximal and 5 cm lateral from the superior border of the patella (15). NIRS-Moxy was enclosed in a black light-blocking case provided by the manufacturer, affixed with adhesive tape, and sampled at 0.5 Hz continuously throughout the entire duration of the exercise. The measurement of $\%S_{mO_2}$ has acceptable reliability and validity across a wide range of exercise intensities (57). A moving average of 0.5 Hz was applied to the $\%S_{mO_2}$ output data and was interpolated at 1-s intervals.

High-intensity exercise capacity via W' Balance modeling. As an index indicating the dynamic fatigue process and its progression to eventual exercise intolerance, W' Balance was computed as previously described (4–6). Because W' Balance is the *net* balance of high-intensity work capacity, a decreasing signal reflects the depletion of energy resources and progression to exhaustion, an increasing signal reflects the repletion of energy resources and distance away from exhaustion, and an approximate zero equates to an imminent limit of exercise tolerance. During exercise performed above CP, W' depletion is directly proportional to the difference between instantaneous power output and CP. When work of exercise is performed below CP, W' repletion depends on the amount of W' remaining and the instantaneous power output relative to CP.

Thus, second-by-second W' (W' Balance) was calculated via the following equation:

$$W'_{\text{Balance}} = W'_0 - W'_{\text{exp}} e^{-\text{Dept}/W'_0}$$

where W'_0 is the initial calculated value of W' from the 3MT, and the W'_{exp} is the expended joules for any work performed above CP (i.e., time of delta CP; D_{Cpt}). W' Balance values (in joules) were converted to relative second-by-second percent by normalizing to the participant's W' (W' Balance %).

Data and statistical analyses. Time series data synchronization across systems occurred through software ‘event’ insertion. Specifically, before data collection, each equipment’s internal device clock was synchronized to a central clock on the collection computer (and thus Windows Time Service). All equipment used to record $\%S_{mO_2}$, $\dot{V}O_2$, and PFC oxygenation record data in time series, with associated time stamps, and the time series data are provided in each data output file. All systems were time aligned on 1-Hz frequency and positioned against the main annotation of ‘exercise start’. NIRS-Moxy monitor was set to directly record at 1 Hz, with the Artinis system (for fNIRS at 25 Hz) and $\dot{V}O_2$ Master (breath by breath) interpolated to 1 Hz given they are recorded

at higher frequency than 1 Hz. Phases of O_2 demand by regional measurement were determined by partitioning the parameters output according to their respective rate of change. Because the primary interest was to identify the segments containing the steepest rate of change for the measurement signal, first, the peak rate of change across the entire 180-s duration of exercise was identified, and a filter to include all values within 50% deviation was applied. Therefore, all values meeting this criterion were defined as “transition phase,” with the boundaries of this phase defining the “onset phase” (all values preceding the transition phase) and the “sustaining phase” (all values following the transition phase). Exercise start equals the 5 s prior to exercise, and exercise end equals the final 5 s of exercise.

All values are reported as means $\pm$ SD unless otherwise stated, and a level of significance was set *a priori* at $P < 0.05$. Effect size was determined using Cohen’s d . Pretesting for parametric statistical tests was performed with Shapiro–Wilk and Brown–Forsythe tests. Specific hypothesis testing examining the start and end of exercise within a measurement variable was performed with Student’s paired t -test. Changes in PFC oxygenation across subregions or hemispheres were compared via one-way ANOVA as was W' Balance across O_2 demand transition phases. Linear regression analyses within the transition phases were performed to examine the relation between O_2 measurement variables and W' Balance. All *post-hoc* examination was performed with Tukey’s test. We excluded one participant who demonstrated an invalid exercise test and failed to meet the criteria of an adequate 3MT. An invalid test was characterized by a flat power output profile from the beginning to the end of the test (indicating evidence of pacing too low of load assignment), and a W' expenditure rate 3 standard deviations above or below the mean. The 3MT power data for this participant met both invalidation criteria. Therefore, the final sample analyzed was a total of 16 participants.

RESULTS

General responses to continuous brief all-out whole-body exercise. Mean 3-min power output was 278 ± 64 W, equating to an average total cumulative work of 49.7 ± 8.6 kJ. End power output was 216 ± 5 W (and defined as CP), and cumulative work above CP was 11.2 ± 3.6 kJ (and defined as W'). From start compared with end exercise, muscle O_2 saturation decreased ($60.1\% \pm 10.8\%$ vs $30.9\% \pm 14.0\%$; $P < 0.0001$, $d = 2.33$) and pulmonary O_2 uptake increased (13.2 ± 4.0 vs 40.8 ± 9.2 mL·kg⁻¹·min⁻¹; $P < 0.0001$, $d = 3.89$). From start compared with end exercise, PFC $\Delta [O_2Hb]$ exhibited a net decrease (-0.04 ± 0.17 vs -3.70 ± 5.27 μM ; $P < 0.0001$, $d = 0.98$), as did PFC $\Delta [Hb \text{ Diff}]$ (-0.01 ± 0.15 vs -7.6 ± 4.79 μM ; $P < 0.0001$, $d = 2.24$), whereas PFC $\Delta [HHb]$ increased (-0.03 ± 0.05 vs 3.85 ± 1.80 μM ; $P < 0.0001$, $d = 3.05$). PFC $\Delta [THb]$ was -0.07 ± 0.20 μM at the start and 0.15 ± 6.25 μM at the cessation of all-out exercise ($d = 0.05$).

General time series patterns across O₂ measurement location to continuous brief all-out whole-body exercise. General time series patterns across O₂ measurement location are displayed in Figure 2. Exercise power output demonstrated the anticipated pattern of an initial peak (414 ± 105 W)

followed by a leveling off in the last 30 s of exercise, with cumulative work performed above CP increasing rapidly from exercise onset but steadily leveling off in the last 30 s of exercise (Fig. 2A). Skeletal muscle O₂ saturation rapidly declined at the start of exercise, followed by a sustained and elongated

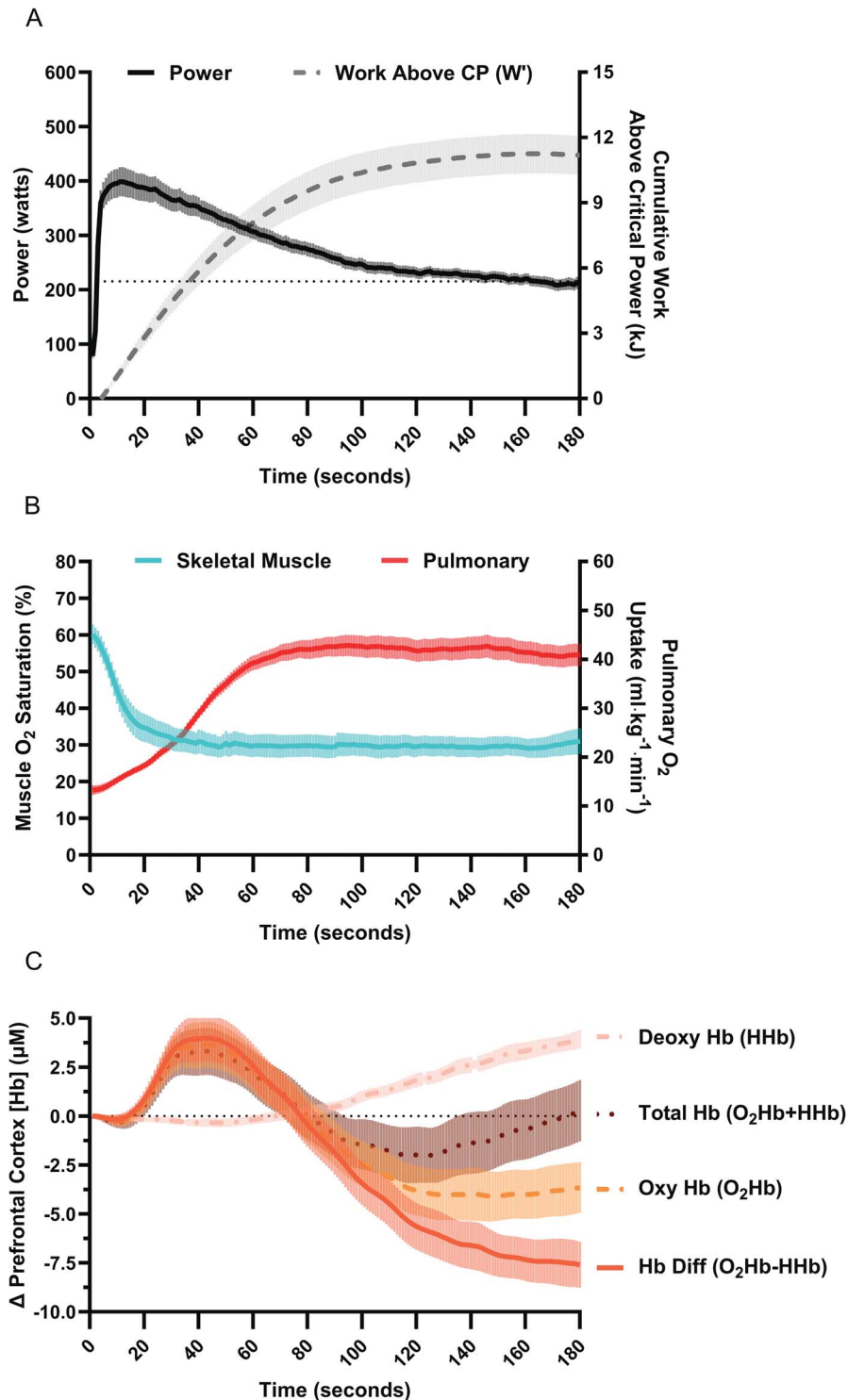


FIGURE 2—General time series patterns across O₂ measurement location to continuous brief all-out whole-body exercise. Time series data for power (black) and W' (gray) expressed as cumulative work performed above CP (A), skeletal muscle O₂ saturation (teal), and pulmonary O₂ uptake (red) (B), and all PFC fNIRS variables (orange hues) across 3 min (expressed in seconds) of continuous all-out cycling exercise. For visual clarity highlighting the dynamic nature of each variable, data are presented as mean $\pm$ SEM ($n = 16$). L, left hemisphere; R, right hemisphere.

nadir. Pulmonary O_2 uptake increased over the first 30–90 s, reaching sustained peak values (Fig. 2B). Global PFC fNIRS indices were as follows. There was a general biphasic pattern across O_2Hb , THb, and Hb Diff, consisting of an initial increase through ~55 s after which a more rapid reduction took place. Following such reductions and through the remaining ~60 s of exercise, Thb directionally increased, O_2Hb stabilized, and Hb Diff continued to decline (Fig. 2C). In contrast, the deoxygenation signal, HHb, was minimally altered through ~55 s, after which it progressively increased until exercise cessation.

PFC fNIRS measurements across subregions and hemispheres during continuous brief all-out whole-body exercise. Time series presentation of subregional and

hemisphere-specific PFC fNIRS output to all-out cycling exercise is displayed in Figures 3A–D. The change from start to end exercise in PFC O_2 mismatch is presented as Hb Diff for all combined subregion and hemisphere signals was $-7.6 \pm 4.7 \mu M$ (Fig. 3E). Combined averages in regional Hb Diff signals across right and left hemispheres were not statistically different from one another (right = $\Delta -7.9 \pm 6.5 \mu M$ vs left = $\Delta -7.2 \pm 3.5 \mu M$; $P = 0.92$, $d = 0.13$). However, for all hemispheres combined in the PFC regions, Hb Diff was significantly lower in the VLPFC compared with the DLPFC or OFC (VLPFC = $\Delta -12.3 \pm 6.7$ vs DLPFC = $\Delta -4.8 \pm 5.3$ or OFC = $\Delta -5.5 \pm 4.7$; $P = 0.0046$, $d = 1.24$, or $P = 0.0016$, $d = 1.18$, respectively). Changes in Hb Diff followed a similar trend regardless of right hemisphere (R VLPFC = Δ

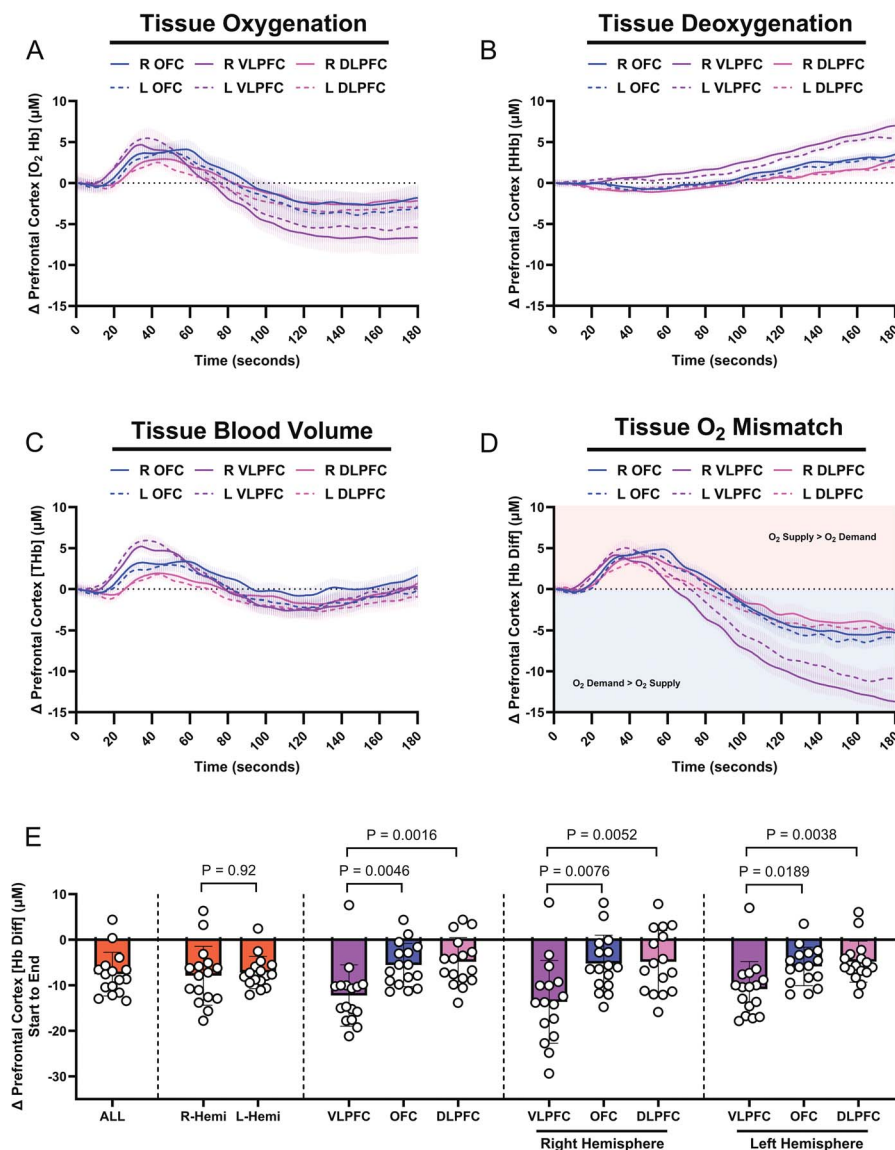


FIGURE 3—fNIRS measurements across PFC brain regions and hemispheres during continuous brief all-out whole-body exercise. Spatiotemporal series data presentation of changes in fNIRS tissue oxygenation [O_2Hb] (A), tissue deoxygenation [HHb] (B), tissue blood volume [THb] (C), and tissue O_2 mismatch [Hb Diff] (D) by region and hemisphere of the PFC (VLPFC, purple; OFC, blue; DLPFC, pink) across the entire 3 min of continuous all-out cycling exercise. For visual clarity highlighting the dynamic nature of each variable, data are presented as mean $\pm$ SEM. Absolute change from start to end exercise (E) is presented for all PFC channels combined (orange), for all regions within right or left hemispheres, across regions independent of hemisphere, and by specific hemisphere within each PFC region. Individual data ($n = 16$) are shown in open circles and mean data $\pm$ SD are shown in bars. P values represent one-way ANOVA for group comparisons across subregions. L, left hemisphere; R, right hemisphere.

-13.6 ± 9.1 vs R DLPFC = $\Delta -4.9 \pm 6.9$ or R OFC = $\Delta -5.2 \pm 6.2$; $P = 0.0052$, $d = 1.08$, or $P = 0.0076$, $d = 1.08$, respectively) or left hemisphere (L VLPFC = $\Delta -10.9 \pm 6.1$ vs L DLPFC = $\Delta -4.8 \pm 4.5$ or L OFC = $\Delta -5.9 \pm 4.2$; $P = 0.0038$, $d = 1.14$, or $P = 0.0189$, $d = 0.96$, respectively).

Sequence of oxygen demand across measurement regions to continuous brief all-out whole-body exercise. Examining the spatiotemporal sequence of O_2 demand signals across body regions of measurement, in which values were normalized and calculated as % change, showed a peak rate of change in skeletal muscle O_2 desaturation (13.8 ± 3.6 s), which was followed next by pulmonary O_2 uptake (44.1 ± 12.3 s), and lastly by PFC O_2 Hb differential (85.0 ± 29.0 s). Note the mathematical inversion for O_2 measurements in skeletal muscle and PFC in order to depict consistent directionality of physiological demand across measurement techniques (Fig. 4A). Time series analysis (see Methods) on the rate of change in the O_2 signal presented three significant and temporally distinct phases within each body region of measurement (Fig. 4B). As follows: 1) an early “onset” phase encompassing a relatively small signal shift that preceded 2) the steepest segment of signal change (labeled as “transition” phase), after which 3) a marked plateau in the signal developed (labeled as “sustaining” phase).

The time at start of transition phase versus start of sustaining phase was significantly different within each region (skeletal muscle = 8.4 ± 9.6 vs 22.4 ± 9.0 s, $P = 0.0139$, $d = 1.50$; pulmonary = 26.4 ± 19.2 vs 60.1 ± 7.9 s, $P < 0.0001$, $d = 2.30$;

PFC = 57.1 ± 20.8 vs 124.2 ± 25.8 s, $P < 0.0001$, $d = 2.86$). Across regions of O_2 measurement, the start of each transition phase occurred in a sequential order from skeletal muscle to pulmonary to PFC (8.4 ± 9.6 to 26.4 ± 19.2 to 57.1 ± 20.8 s, respectively; skeletal muscle vs pulmonary was $P = 0.0006$, $d = 1.19$, skeletal muscle vs PFC was $P < 0.0001$, $d = 3.00$, pulmonary vs PFC was $P < 0.0001$, $d = 1.53$). Similarly, the start of each sustaining phase (i.e., end of each transition phase) occurred in a sequential order from skeletal muscle to pulmonary to PFC (22.4 ± 9.0 to 60.1 ± 7.9 to 124.2 ± 25.8 s, respectively; skeletal muscle vs pulmonary was $P < 0.0001$, $d = 4.45$, skeletal muscle vs PFC was $P < 0.0001$, $d = 5.27$, pulmonary vs PFC was $P < 0.0001$, $d = 3.36$) (Fig. 4B). The transpiring times during exercise at which the end of the transition phase for skeletal muscle adjoined with pulmonary start of the transition phase, as well as the end of the transition phase for pulmonary and PFC start of the transition phase, were not significantly different ($P = 0.927$, $d = 0.27$, and $P = 0.976$, $d = 0.19$, respectively). These relationships are further depicted in Figures 4C–D. Specifically observed during all-out exercise, muscle O_2 desaturation increases predominantly in the absence of change in pulmonary O_2 uptake wherein following the noted transition phase for S_mO_2 , pulmonary O_2 uptake increases predominantly in the absence of change in muscle O_2 desaturation (Fig. 4C). Similarly, both muscle O_2 desaturation and pulmonary O_2 uptake increased most in the absence of change in PFC O_2 Hb differential, whereupon following the noted transition phases within both muscle and pulmonary regions, the

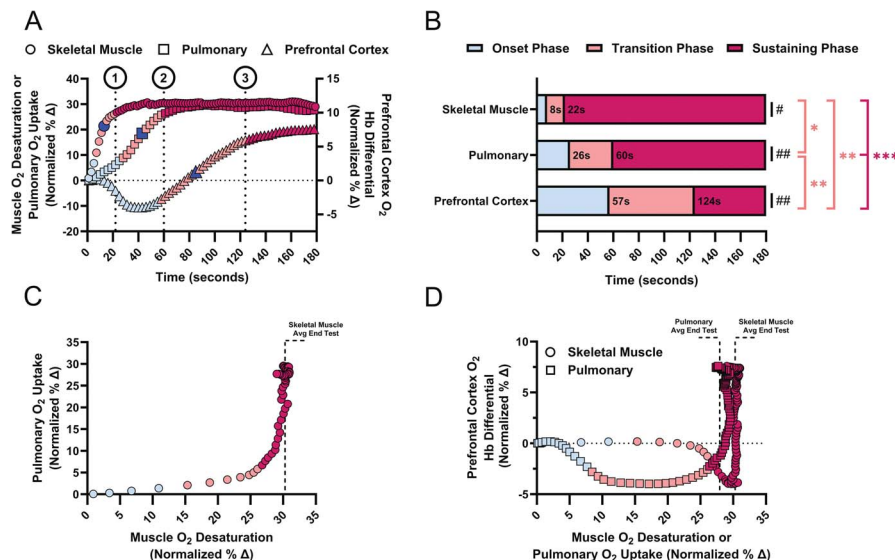


FIGURE 4—Sequence across regional indices of oxygen demand to continuous brief all-out whole-body exercise: indications for region-specific transitions and maximums. By-region time series indices of O_2 demand measured at the skeletal muscle, pulmonary, and PFC (all subregions and hemispheres combined) across 3 min (expressed in seconds) of continuous all-out cycling exercise (A). Phases per tissue regional measurement by time for differing rate of change, with an inset of seconds reflecting the specific time point identified as the boundary between phases (B). Average relation in measurement of skeletal muscle to pulmonary (C). Average relation in measurement of skeletal muscle or pulmonary to PFC (D). Graphical details are as follows: skeletal muscle, open circles; pulmonary, open squares; PFC, open triangles; onset phase, light blue; transition phase, light orange; sustaining phase, dark red; peak rate of change, dark blue symbol; encircled number 1, end skeletal muscle transition phase; encircled number 2, end pulmonary transition phase; encircled number 3, end PFC transition phase. Note that muscle and PFC measurements are inversely presented to align with pulmonary O_2 uptake. Data are normalized to % change of 100 units to account for differing tissue region-specific measurement tools and are expressed as mean for visual clarity ($n = 16$). Statistical comparison performed by two-way ANOVA with # as $P = 0.0139$ and ## as $P < 0.0001$ for within that given region across start and end transition phase. Transition phase start time across regions was * as $P = 0.0006$ and ** as $P < 0.0001$. *** also represents $P < 0.0001$; however, for transition phase end time across all regions.

change in Hb differential increased greatly despite no further notable changes in muscle or pulmonary indices (Fig. 4D).

Exercise tolerance and expenditure of $\dot{W}$ during continuous brief all-out whole-body exercise are sequentially related to peripheral-to-central oxygen dependency. Three-minute all-out exercise evoked complete depletion of $\dot{W}$ (average end test $\dot{W}$ Balance = $-3.1\% \pm 1.6\%$), consistent with maximal depletion of work capacity above CP and participant exhaustion. Continuous mean signals of skeletal muscle O_2 saturation, pulmonary O_2 uptake, and PFC Hb Diff are presented over progressive reduction in $\dot{W}$ throughout exercise in Figure 5A. Figure 5B shows coinciding values of $\dot{W}$ Balance with the end transition phase across body regional O_2 measurement and depicts a progressive shift across regions in O_2 limitation from skeletal muscle to pulmonary to PFC until a $\dot{W}$ Balance of about zero is reached and no further change in O_2 measurement variables takes place. $\dot{W}$ Balance through the transition phase was very closely correlated with skeletal muscle O_2 desaturation ($R^2 = 0.910$), pulmonary O_2 uptake ($R^2 = 0.997$), and PFC O_2 Hb Diff ($R^2 = 0.968$) (Fig. 5C). Values of $\dot{W}$ Balance were neither significantly different at the end of the transition phase for skeletal muscle ($71.1\% \pm 13.6\%$) compared with the start of the transition phase for pulmonary ($64.6\% \pm 17.1\%$; $P = 0.56$, $d = 0.42$), or the end of the transition phase for pulmonary ($25.9\% \pm 10.3\%$) compared with the

start of transition phase for PFC ($30.4\% \pm 11.8\%$; $P = 0.83$, $d = 0.41$) (Fig. 5D). However, $\dot{W}$ Balance values were significantly different within the transition phase (i.e., at the start vs the end of the transition phase; $P < 0.0001$; Fig. 5D).

DISCUSSION

The purpose of this study was to investigate whether the expended rate of work above CP ($\dot{W}$ Balance) during all-out whole-body exercise is related to a decline in PFC oxygenation secondary to an organized systemic outstripping of muscle O_2 supply relative to O_2 demand. All-out exercise was used to establish a restrained O_2 supply to working muscle that was coupled to the exhaustion of high-energy resources and work capacity above CP. Accordingly, a significant O_2 mismatch favoring an O_2 demand greater than O_2 supply developed within multiple PFC regions across both hemispheres during all-out exercise. This followed a sequence of interconnected *a priori* events: 1) progression toward maximal O_2 delivery and extraction in active skeletal muscle, 2) progression toward maximal whole body O_2 consumption, and 3) progressive expenditure of available energy resources at work rates above CP known as the high-intensity work capacity $\dot{W}$. Moreover, the largest rate of change in O_2 demand (termed transition phase) prior to leveling off at peak values (termed sustaining phase)

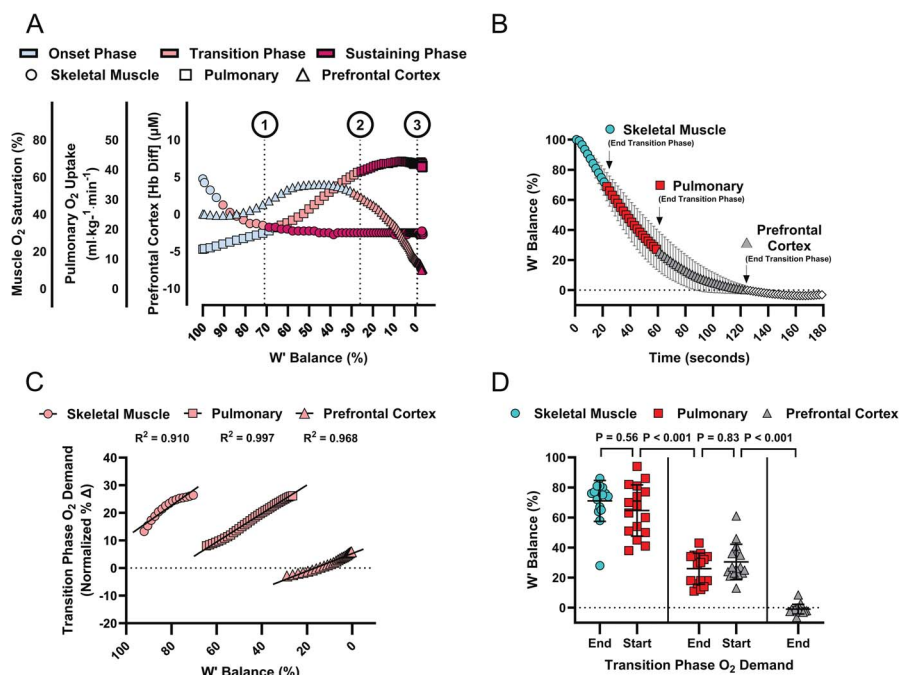


FIGURE 5—Depletion of high-intensity work capacity and O_2 demand across tissue regions during continuous brief all-out whole-body exercise. Values of skeletal muscle O_2 saturation, pulmonary O_2 uptake, or PFC O_2 mismatch (all subregions combined) across $\dot{W}$ Balance, and presentation of the identified O_2 demand phases (in color) during continuous all-out cycling exercise (A). Time series of $\dot{W}$ Balance with color-denoted regions of O_2 measurement for the nonsustaining phases (i.e., up to end transition phase) (B). Linear regression between regional O_2 measurement and $\dot{W}$ Balance, specifically for the transition phase (C). Individual and mean $\dot{W}$ Balance values occurring at the start and end transition phases by region (D). Graphical details are as follows: skeletal muscle, open circles, teal color; pulmonary, open squares, red color; PFC, open triangles, gray color; onset phase, light blue; transition phase, light orange; sustaining phase, dark red; encircled number 1, end skeletal muscle transition phase; encircled number 2, end pulmonary transition phase; encircled number 3, end PFC transition phase. Panel A presents data as mean for visual clarity, panel B presents data as mean $\pm$ SD, panel C presents data as time series mean expressed as normalized to % change of 100 units to account for differing region-specific measurement tools ($n = 16$). Linear regression analysis determined R^2 values. Statistical comparison performed by one-way ANOVA with P values for different $\dot{W}$ within end-transition and start-transition phases, respectively.

exposed an O₂ mismatch cascade of O₂ supply relative to O₂ demand that started peripherally and ended centrally (skeletal muscle → pulmonary → PFC) in accordance with the magnitude of *W'* depletion. Taken together, although these data cannot provide mechanisms of causation, they generally suggest the potential for an integrative and sequential progression of O₂ mismatch from skeletal muscle to brain that may underpin the limits of tolerance to all-out maximal whole-body exercise in humans.

Oxygenation of the PFC during whole-body all-out exercise: Subregion and hemisphere specific insight.

To better understand the oxygenation profile across subregions and hemispheres of the PFC in the brain during exhaustive all-out whole-body exercise, we measured continuous changes in the concentration of oxy- and deoxyhemoglobin chromophores (O₂Hb and HHb, respectively), and calculated indices of tissue blood volume (THb) and O₂ mismatch ($p_{\text{PFC}}\text{O}_2$). Globally across the PFC, all-out exercise evoked an initial and simultaneous increase in the fNIRS oxygenation signal, which was equally met with an increase in tissue blood volume. With an unchanged deoxygenation signal, this resulted in a positive Hb differential. These data are consistent with a tissue O₂ matching signature that favors a larger O₂ supply than metabolic demand of the PFC during the first minute of all-out exercise (Fig. 2C). In direct contrast, during the second minute of all-out exercise, O₂Hb and THb both began to steeply decline, while HHb began a progressive rise, thus indicating a marked imbalance in O₂ demand relative to O₂ supply. Over the third and final minute of exercise prior to exhaustion, PFC Hb differential continued to decrease, largely due to an increase in HHb, highlighting an expanded PFC O₂ mismatch.

These data are directionally consistent with the report by Robertson and Marino showing a steep decline in brain electrical activity of the VLPFC at an exercise intensity above the respiratory compensation point (i.e., approximately CP), and secondary evidence showing an overall decline in cerebral tissue oxygenation measured via NIRS prior to maximal power output in cycle ramp tests (14,15,33,58). Historical documentation during graded intensity protocols often suggests that PFC deoxygenation occurs only in conjunction with work rates considered severe-intensity exercise but not during heavy-intensity exercise (15,16). A biphasic pattern in brain oxygenation is not uncommon, with various reports showing increases in oxygenation during exercise, and some observing a decline at very high intensity (14,15,33,58). Thus, it could be argued that our observations are simply: 1) compressed and 2) reversed, compared with progressive ramped max exercise protocols, given the nature of the 3MT exercise protocol starting at maximal all-out effort. Our data, however, may illustrate that it is not (severe domain) high-intensity work rate *per se* that drives a lowering of PFC oxygenation indices during maximal exercise, but rather the corresponding underlying processes, because no reduction in PFC O₂ indices was observed the entire first one-third of exercise when peak power output and the largest portion of work performed above CP occurred.

We utilized a comprehensive 24-channel fNIRS arrangement to widely investigate oxygenation profiles across the

PFC and determine the potential presence of any O₂ mismatch in specific subregions during all-out exercise. This rationale developed in part because severe-intensity and/or exhaustive exercise has previously suggested the potential for independent PFC oxygenation profiles across subregions and hemispheres (20,24,33,34), because historical studies have often been regionally nonspecific or constrained with their fNIRS measurements (12,14,15,34,58), and because psychological and behavioral outcomes are brain region specific (25,27,29). As shown in Figures 3D and E, continuous all-out whole-body exercise appears to elicit region-specific differences in the magnitude of O₂ mismatch, with a larger exercise-mediated O₂ Hb differential occurring in the VLPFC versus the OFC or DLPFC (Fig. 3E). These findings appear to corroborate that underlying changes across regions reflect different ongoing neural processes during the actual all-out exercise itself, even though brain functions are not isolated within a given region, and during any given task, multiple large-scale networks are functioning together (59). Interestingly, although the PFC is in general, paramount in regulating our thoughts, decisions, emotions, and actions, the VLPFC region is engaged during task reappraisal in which negative emotions may undergo reevaluation, as well as during events involving emotional suppression (29). Notably, our findings show the greatest O₂ Hb differential occurred in the VLPFC (Fig. 3). This could have implications for exercise tolerance because suppressing the sense of fatigue and feelings of displeasure (19,21,33) becomes more difficult as oxygen availability to neurons decreases and may thereby exacerbate the fatigue response due to lack of VLPFC inhibition of these feelings. The left VLPFC has a role in selecting a semantic reappraisal of a situation, and the right VLPFC aims to buffer negative thoughts as well as override motor responses to stop a task in the presence of uncertainty (60). Indeed, general participant feedback in the current study regularly described the exercise protocol we deployed (i.e., 3 min of continuous all-out effort) as one of, if not *the*, most overwhelming and perturbing voluntary activities. Although to a lesser extent than the VLPFC, but still notably significant, a large O₂ Hb difference was also observed in the OFC, which is known to aid in the process of decision making. The smallest O₂ imbalance (favoring less *net* Hb differential) was observed in the DLPFC, and although any mechanistic reasoning cannot be defined by the present study, maintenance of tissue oxygenation may not be unsurprising given a suggested key role for the right DLPFC in sustaining attention and high alertness to threat cues (26), including homeostatic perturbations within the body (27,29,31). Collectively, although extrapolation extends beyond the scope of the present study, our physiological data appear in line with the broad psychosocial understanding of acute stress-mediated impairments in PFC function and begin to highlight that continuous maximal “exercise stress” may be of equal potency (61).

Spatiotemporal patterns and the integrative control during systemic oxygen mismatch. All-out exercise appeared to evoke a distinguished spatiotemporal pattern of O₂ mismatch across regions and hemispheres. To our knowledge, no studies to date have performed multiregion, high-frequency continuous NIRS recordings across both hemispheres of the

PFC during all-out exercise, and only few studies (17) have examined and reported declines PFC oxygenation during continuous severe-intensity exercise to exhaustion. These findings, along with other studies examining PFC oxygenation during incremental exercise, have led to the general conclusion that the response is dependent on the exercise intensity wherein oxygenation increases occur prior to intense exercise and decrease as exercise intensity is heightened. The data from the present study are the opposite: greatest PFC oxygenation during peak power outputs and lowest during lesser power output, thus ultimately highlighting the importance to consider what physiological perturbations an exercise protocol will induce. Our study suggests that inspecting the spatiotemporal second-by-second response in PFC O₂ Hb differential reveals a clear biphasic response pattern, initially increasing in support of O₂ delivery, prior to a robust shift toward an overweighted O₂ demand versus O₂ supply. This finding highlights two important points. First, analyzing changes in PFC O₂ on a second-by-second basis during all-out exercise may reveal additional insights into dynamic patterns, whereas employing larger epoch averaging may omit underlying dynamics. Second, the effect of exercise on PFC oxygenation may be less dependent on the absolute intensity of exercise *per se*, but rather on the magnitude of metabolic perturbations driven by sequential limitation in peripheral O₂ supply concomitant to the depletion of $\dot{V}O_2$. Quantifying exercise intensity based on external workload versus the metabolic responses it elicits continues to be debated (62). Our data indicate that peak power output, reached within the first ~12 s of exercise, did not temporally coincide with peak values in O₂ indices collected either in locally active skeletal muscle, at the lungs for whole body estimates, or across subregions and hemispheres in the PFC. Instead, these peripheral and central factors responded sequentially, reaching their maximum (sustaining phase) only near the end of the exercise trial duration, when power output was lowest, yet simultaneously the point at which metabolic perturbations are highest.

Continuous all-out whole-body cycling exercise evokes: (a) maximal whole-body oxygen consumption, which originates in part from the involvement of very large muscle mass recruitment and the robust increase in skeletal muscle metabolism, (b) maximal cardiac output and muscle O₂ extraction, and (c) peripheral vasodilation and blood flow redistribution great enough to compromise blood pressure (7,10,12,18). Because of this, we aimed to explore the coordinated relationship between local cerebral O₂ demand in the PFC versus indices of O₂ demand across local skeletal muscle and at the whole-body level. Here, we reveal a sequential pattern of peripheral-to-central O₂ responses, featuring three distinct but common phases across each location of O₂ measurement (i.e., skeletal muscle, pulmonary, and PFC). The universal pattern is as follows: 1) an initial onset phase with a minor shift in signal followed by 2) a transition phase featuring the steepest rate of change in the respective signal, leading to 3) a sustaining phase where each signal approaches a leveling off at its corresponding peak or nadir. Interestingly, the timing of each phase progresses sequentially from skeletal muscle to pulmonary/circulatory to PFC in a manner that appears to coordinate the onset and offset

phases. For example, as shown in Figures 4A–B, these data indicate skeletal muscle is the first to desaturate, quickly progressing through its onset phase and proceeding through the sustaining phase between ~8 and ~22 s, a time at which pulmonary O₂ uptake is entering its onset phase. Not until skeletal muscle O₂ desaturation approaches flatline and enters its sustaining phase at ~22 s does pulmonary O₂ uptake rapidly increase (Fig. 4C), beginning its steepest rate of rise and thus transition phase (~26 to ~60 s). With pulmonary O₂ uptake continuing to rise rapidly to match the metabolic O₂ demand of exercising muscle, the PFC O₂ response continues to increase in its onset phase. However, as pulmonary O₂ uptake reaches a maximum with no further ability to increase (18) and enters its sustaining phase (~60 s), only then does PFC O₂ Hb differential (~57 s) begin a reversal into a steep rate of change in O₂ demand (Fig. 4A) and display the greatest magnitude of change (Fig. 4D). This demarcation in the PFC transition phase appears to occur simultaneously with skeletal muscle and pulmonary O₂ indices at their respective maximum sustaining phase (Fig. 4D). Finally, around ~124 s, as $\dot{V}O_2$ Balance approaches its nadir (Fig. 5B) and power output approaches a plateau at the highest sustainable power (i.e., CP) (Fig. 2A), does PFC O₂ Hb differential enter its sustaining phase until exercise termination (Figs. 4A–B).

We interpret the aforementioned spatiotemporal patterns to indicate that an O₂ mismatch of insufficient O₂ supply relative to O₂ demand occurring in the PFC may result secondarily to the systemic outstripping of O₂ supply and be particularly compromised during all-out exercise that continues long enough to evoke significant metabolic perturbations of O₂ mismatch within skeletal muscle. Such observation would further highlight the potency of skeletal muscle as ‘the sleeping giant’ (63); however, it is equally important to remember that O₂ Hb affinity is not fixed but also modulated by dynamic factors concurrent with exercise intensity such as temperature and acidity. During all-out exercise, skeletal muscle is fully recruited (7), muscle vasodilation exceeds cardiac output (11), and whole-body O₂ uptake is forced to stay at maximum values (18). At least initially, systemic hemodynamic regulation and the control of O₂ delivery during continuous all-out exercise may serve to increase oxygenation to the PFC during corresponding desaturation of skeletal muscle. However, this preservation of (or over supply of) PFC oxygenation lasts only up through to the process of matched *systemic* O₂ supply and demand, but not further, for when whole body O₂ uptake and underlying factors are near maximal (i.e., cardiac output and O₂ extraction), oxygenation to the PFC appears threatened and begins to decline. A flatline signal occurring after a rapid rate of change in muscle O₂ saturation and whole-body O₂ uptake could indicate that equilibrium has been restored and O₂ supply meets demand, or it could indicate that O₂ supply has reached a point where it cannot increase and cannot meet the metabolic demand (i.e., too great of an O₂ mismatch). Given that continuous all-out exercise was prescribed, a maintenance or increase in power output would be expected in the former scenario, whereas if the latter circumstances were present, power output

would be assumed to fall to a critical balancing point as a functional means to reduce metabolic demand and stabilize mechanisms of O_2 supply. Interestingly, although we do not have measurements to indicate the magnitude of arterial hypoxemia in the present study, the sequential kinetics of tissue oxygenation on peripheral and central limits of power output and exercise tolerance align well with the schema proposed by Amann and Dempsey suggesting peripheral-mediated fatigue up to the point at which either cerebral oxygenation becomes compromised or muscle acidosis overwhelms and motoneuronal output suppression ensues (35,36).

Progressive depletion of W' is associated with the sequence of systemic oxygen mismatch. The capacity to perform work above CP and the concurrent proximity to volitional exhaustion has been demonstrated via calculation of W' (and D' equivalent) (2–6,10); however, the mechanistic understanding of W' continues to remain indistinct with no examination to date exploring a potential role of the brain. With spatiotemporal analysis by O_2 measurement location revealing a transition phase characterized by the steepest rate of change followed by a maximal sustaining plateau, and a progression of this pattern occurring sequentially first at the muscle (~14 s) → pulmonary (~44 s) → brain (~85 s) (Figs. 4A–B), we sought to understand whether this pattern could be explanatory of W' Balance during all-out exercise (Fig. 5A). Indeed, the transition phase O_2 indices were strongly correlated with expenditure of W' Balance (muscle, $R^2 = 0.91$; pulmonary, $R^2 = 0.997$; PFC, $R^2 = 0.968$; Fig. 5C), with crossover between regions occurring at the same W' Balance and with the end of the PFC O_2 transition phase occurring at complete depletion of W' and a settling of work rate at approximate CP (Figs. 5B, D). Hints of a relationship between W' Balance and matching of O_2 supply and demand have been reported (6,8,10). That said, the present investigation provides original evidence for 1) key sequence of events involving systemic progression of O_2 mismatch across tissue regions 2) a role for the PFC in the expenditure of W' , especially given a coinciding maximal plateau/sustaining phase in PFC O_2 Hb differential and the complete depletion of W' (Fig. 5B). The latter of which is usually reserved for the point of exercise cessation, a suppression of maximal work rate to only near CP, or a ceiling of effort and moods state that confers with very significant discomfort (and as we describe above, a significant reduction in VLPFC O_2). More specifically, the magnitude of the O_2 mismatch highlighted in this study may help elucidate, at least partially, the extent of discomfort (physical and psychological) experienced during such all-out testing compared with other, more gradual maximal tests (e.g., graded exercise testing).

Experimental considerations. The data provided cannot be used to conclude mechanisms of causation. We used a single reverse intensity gradation cycle ergometry protocol approach to maximal exercise (i.e., brief continuous all-out exercise characterized by a rapid peak and decline in maximal power output). Although not the intent of the present study, no secondary validation condition was employed by making an independent estimate of responses to severe- or heavy-intensity exercise, meaning

above or below CP, respectively. More mechanistically, PFC control of tissue oxygenation has a close relation to arterial CO_2 at rest and during exercise, and such changes may very well be closely linked to arterial or end-tidal CO_2 indices (58). The present study, however, did not examine these variables; thus, future study is ripe to address such questions and any interrelation to W' expenditure and high-intensity work capacity.

Another consideration and one of historical debate is on the extent of superficial versus deeper tissue contributions given that NIRS measures arbitrary units of light-absorbing chromophores under the path length of only that sensor (45). Although we do not interpret the potential of Mayer wave oscillations to change the overarching conclusions offered in this report, they can be present in fNIRS signals but were not specifically addressed or studied in our investigation. Regarding the PFC specifically, a contribution of extracerebral oxygenation (e.g., skin) within the fNIRS signal is possible, albeit minimal during even clamping of the internal carotid artery (64) as well as less extreme maneuvers (65). We observed an increase in [HHb] suggesting changes largely independent of a skin flow response (66). The consideration is furthered by reports showing that forehead Doppler flux and [HHb] are poorly correlated during exercise (67), nor are explanatory during blood pressure challenge (68). Moreover, facial skin blood flow responses are presumed to be universal during exercise (69), rather than present distinct subregional location-specific differences like we observed (Fig. 3). Lastly, forehead cutaneous vascular responses are most strongly correlated with external carotid artery blood flow, which increases during intense exercise, where in contrast, cerebral oxygenation is more largely influenced by internal carotid artery blood flow, which is seen to decrease during intense exercise and is consistent with presented data showing a net decline in PFC oxygenation (64,70). Although we are confident to conclude that PFC O_2 Hb differential does decline during all-out exercise, and that an O_2 mismatch favoring an insufficient O_2 supply relative to O_2 demand is present, whether that resulted via suppressed neuronal activity, depressed blood flow, or any number of combinations of a mismatch between O_2 supply and O_2 demand in the PFC during maximal all-out exercise warrants further investigation.

We also recognize the lack of direct measures for cardiac output, mean arterial pressure, or global leg or brain blood flow. That said, a case for the more first order tissue oxygen measures used in the present study could also be made. Methodological collection of such hemodynamic variables during continuous all-out whole body cycling exercise should not be underestimated, especially in the context of examining spatiotemporal patterns that require dynamic uninterrupted times series data presentation. Invasive methods (i.e., catheterization) for cardiac output and active muscle (leg) blood flow (such as thermodilution) during all-out exercise are in theory plausible; however, the collection of all variables *simultaneously* and in time series during high movement “prolonged sprint” cycling is not trivial, with much complexity and nuance. Nevertheless, the combined approach in measuring local skeletal muscle O_2 saturation, pulmonary O_2 uptake, and PFC

oxygenation during exercise does provide the data frequency (1 Hz or better) and primary variable type (end organ tissue oxygen indices) desired, while still helping to elucidate the integrative challenge of matching O₂ supply to metabolic demand during all-out whole-body exercise.

CONCLUSIONS

Continuous brief all-out cycling exercise elicits a transient increase in PFC oxygenation; however, in simultaneous conjunction with the onset of maximal deoxygenation of skeletal muscle and maximal whole-body pulmonary O₂ uptake, this signal reverses and begins a marked decline toward a clear O₂ mismatch of O₂ supply < O₂ demand. High-intensity work capacity performed above CP and defined as the magnitude of $\dot{W}$ Balance depleted appears linked to physiological underpinnings comprised of an initial local skeletal muscle deoxygenation, a subsequent increase in systemic whole-body O₂ consumption, and subsequent declines in region specific PFC oxygenation.

REFERENCES

1. Vanhatalo A, Doust JH, Burnley M. Determination of critical power using a 3-min all-out cycling test. *Med Sci Sports Exerc.* 2007;39(3):548–55.
2. Poole DC, Burnley M, Vanhatalo A, Rossiter HB, Jones AM. Critical power: an important fatigue threshold in exercise physiology. *Med Sci Sports Exerc.* 2016;48(11):2320–34.
3. Vanhatalo A, Black MI, DiMenna FJ, et al. The mechanistic bases of the power–time relationship: muscle metabolic responses and relationships to muscle fibre type. *J Physiol.* 2016;594(15):4407–23.
4. Kirby BS, Winn BJ, Wilkins BW, Jones AM. Interaction of exercise bioenergetics with pacing behavior predicts track distance running performance. *J Appl Physiol (1985).* 2021;131(5):1532–42.
5. Skiba PF, Clarke D, Vanhatalo A, Jones AM. Validation of a novel intermittent $\dot{W}$ model for cycling using field data. *Int J Sports Physiol Perform.* 2014;9(6):900–4.
6. Skiba PF, Chidnok W, Vanhatalo A, Jones AM. Modeling the expenditure and reconstitution of work capacity above critical power. *Med Sci Sports Exerc.* 2012;44(8):1526–32.
7. Vanhatalo A, Poole DC, DiMenna FJ, Bailey SJ, Jones AM. Muscle fiber recruitment and the slow component of O₂ uptake: constant work rate vs. all-out sprint exercise. *Am J Physiol Regul Integr Comp Physiol.* 2011;300(3):R700–7.
8. Broxterman RM, Ade CJ, Craig JC, Wilcox SL, Schlup SJ, Barstow TJ. Influence of blood flow occlusion on muscle oxygenation characteristics and the parameters of the power–duration relationship. *J Appl Physiol (1985).* 2015;118(7):880–9.
9. Broxterman RM, Craig JC, Smith JR, et al. Influence of blood flow occlusion on the development of peripheral and central fatigue during small muscle mass handgrip exercise. *J Physiol.* 2015;593(17):4043–54.
10. Kirby BS, Clark DA, Bradley EM, Wilkins BW. The balance of muscle oxygen supply and demand reveals critical metabolic rate and predicts time to exhaustion. *J Appl Physiol (1985).* 2021;130(6):1915–27.
11. Hammer SM, Alexander AM, Didier KD, Huckaby LM, Barstow TJ. Limb blood flow and muscle oxygenation responses during handgrip exercise above vs. below critical force. *Microvasc Res.* 2020;131:104002.
12. González-Alonso J, Dalsgaard MK, Osada T, et al. Brain and central haemodynamics and oxygenation during maximal exercise in humans. *J Physiol.* 2004;557(1):331–42.
13. Orcioli-Silva D, Beretta VS, Santos PCR, et al. Cerebral and muscle tissue oxygenation during exercise in healthy adults: a systematic review. *J Sport Health Sci.* 2024;13(4):459–71.
14. Rupp T, Perrey S. Prefrontal cortex oxygenation and neuromuscular responses to exhaustive exercise. *Eur J Appl Physiol.* 2008;102(2):153–63.
15. Subudhi AW, Miramon BR, Granger ME, Roach RC. Frontal and motor cortex oxygenation during maximal exercise in normoxia and hypoxia. *J Appl Physiol (1985).* 2009;106(4):1153–8.
16. Jung R, Moser M, Baucsek S, Derm S, Schneider S. Activation patterns of different brain areas during incremental exercise measured by near-infrared spectroscopy. *Exp Brain Res.* 2015;233(4):1175–80.
17. Shibuya K, Tanaka J, Kuboyama N, Murai S, Ogaki T. Cerebral cortex activity during supramaximal exhaustive exercise. *J Sports Med Phys Fitness.* 2004;44(2):215–9.
18. Burnley M, Doust JH, Vanhatalo A. A 3-min all-out test to determine peak oxygen uptake and the maximal steady state. *Med Sci Sports Exerc.* 2006;38(11):1995–2003.
19. Ekkekakis P. Illuminating the black box: investigating prefrontal cortical hemodynamics during exercise with near-infrared spectroscopy. *J Sport Exerc Psychol.* 2009;31(4):505–53.
20. Robertson CV, Marino FE. A role for the prefrontal cortex in exercise tolerance and termination. *J Appl Physiol (1985).* 2016;120(4):464–6.
21. Bigliassi M, Filho E. Functional significance of the dorsolateral prefrontal cortex during exhaustive exercise. *Biol Psychol.* 2022;175:108442.
22. Hartman ME, Ekkekakis P, Dicks ND, Pettitt RW. Dynamics of pleasure–displeasure at the limit of exercise tolerance: conceptualizing the sense of exertional physical fatigue as an affective response. *J Exp Biol.* 2019;222(3):1–10.
23. Venhorst A, Micklewright DP, Noakes TD. The psychophysiological determinants of pacing behaviour and performance during prolonged endurance exercise: a performance level and competition outcome comparison. *Sports Med.* 2018;48(10):2387–400.
24. Tempest G, Parfitt G. Self-reported tolerance influences prefrontal cortex hemodynamics and affective responses. *Cogn Affect Behav Neurosci.* 2016;16(1):63–71.
25. Spielberg JM, Stewart JL, Levin RL, Miller GA, Heller W. Prefrontal cortex, emotion, and approach/withdrawal motivation. *Soc Pers Psychol Compass.* 2008;2(1):135–53.
26. Mannarelli D, Pauletti C, Grippo A, et al. The role of the right dorso-lateral prefrontal cortex in phasic alertness: evidence from a contingent negative variation and repetitive transcranial magnetic stimulation study. *Neural Plast.* 2015;2015:410785.
27. Figner B, Knoch D, Johnson EJ, et al. Lateral prefrontal cortex and self-control in intertemporal choice. *Nat Neurosci.* 2010;13(5):538–9.
28. Tanji J, Hoshi E. Behavioral planning in the prefrontal cortex. *Curr Opin Neurobiol.* 2001;11(2):164–70.

Our data suggest an integrative and sequential progression of O₂ mismatch from skeletal muscle to the brain may underpin the limits of exercise tolerance during all-out maximal whole-body exercise in humans.

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29. Goldin PR, McRae K, Ramel W, Gross JJ. The neural bases of emotion regulation: reappraisal and suppression of negative emotion. *Biol Psychiatry*. 2008;63(6):577–86.
30. Ochsner KN, Silvers JA, Buhle JT. Functional imaging studies of emotion regulation: a synthetic review and evolving model of the cognitive control of emotion. *Ann N Y Acad Sci*. 2012;1251(1):E1–24.
31. Lind E, Welch AS, Ekkekakis P. Do “mind over muscle” strategies work? Examining the effects of attentional association and dissociation on exertional, affective and physiological responses to exercise. *Sports Med Auckl N Z*. 2009;39(9):743–64.
32. Commentaries on Viewpoint: a role for the prefrontal cortex in exercise tolerance and termination. *J Appl Physiol* (1985). 2016;120(4):467–9.
33. Robertson CV, Marino FE. Prefrontal and motor cortex EEG responses and their relationship to ventilatory thresholds during exhaustive incremental exercise. *Eur J Appl Physiol*. 2015;115(9):1939–48.
34. Tempest GD, Eston RG, Parfitt G. Prefrontal cortex haemodynamics and affective responses during exercise: a multi-channel near infrared spectroscopy study. *PLoS One*. 2014;9(5):e95924.
35. Amann M, Dempsey JA. Ensemble input of group III/IV muscle afferents to CNS: a limiting factor of central motor drive during endurance exercise from normoxia to moderate hypoxia. *Adv Exp Med Biol*. 2016;903:325–42.
36. Hureau TJ, Broxterman RM, Weavil JC, Lewis MT, Layec G, Amann M. On the role of skeletal muscle acidosis and inorganic phosphates as determinants of central and peripheral fatigue: a 31 P-MRS study. *J Physiol*. 2022;600(13):3069–81.
37. Dicks ND, Jamnick NA, Murray SR, Pettitt RW. Load determination for the 3-minute all-out exercise test for cycle ergometry. *Int J Sports Physiol Perform*. 2016;11(2):197–203.
38. Johnson TM, Sexton PJ, Placek AM, Murray SR, Pettitt RW. Reliability analysis of the 3-min all-out exercise test for cycle ergometry. *Med Sci Sports Exerc*. 2011;43(12):2375–80.
39. Colier WJNM, Van Haaren NJ, Oeseburg B. A comparative study of two near infrared spectrophotometers for the assessment of cerebral haemodynamics. *Acta Anaesthesiol Scand*. 1995;39(s107):101–5.
40. Hong Y, Bao D, Manor B, Zhou Y, Zhou J. Effects of endurance exercise on physiologic complexity of the hemodynamics in prefrontal cortex. *Neurophotonics*. 2024;11(1):015009–9.
41. Li SN, Anbalagan P, Pang J, Ihsan M, Girard O. Exercise responses to repeated cycle sprints with continuous and intermittent hypoxic exposure. *Eur J Sport Sci*. 2024;ejsc.12146.
42. Van Beekvelt MCP, Van Engelen BGM, Wevers RA, Colier WJNM. *In vivo* quantitative near-infrared spectroscopy in skeletal muscle during incremental isometric handgrip exercise. *Clin Physiol Funct Imaging*. 2002;22(3):210–7.
43. Grassi B, Quaresima V, Marconi C, Ferrari M, Cerretelli P. Blood lactate accumulation and muscle deoxygenation during incremental exercise. *J Appl Physiol* (1985). 1999;87(1):348–55.
44. Scholkmann F, Kleiser S, Metz AJ, et al. A review on continuous wave functional near-infrared spectroscopy and imaging instrumentation and methodology. *Neuroimage*. 2014;85(Pt 1):6–27.
45. Barstow TJ. Understanding near infrared spectroscopy and its application to skeletal muscle research. *J Appl Physiol* (1985). 2019;126(5):1360–76.
46. Sharbrough F, Chatrjian GE, Lesser R, Luders H, Nuwer M, Picton T. American Electroencephalographic Society guidelines for standard electrode position nomenclature. *Clin Neurophysiol*. 1991;8:200–2.
47. Homan RW, Herman J, Purdy P. Cerebral location of international 10–20 system electrode placement. *Electroencephalogr Clin Neurophysiol*. 1987;66(4):376–82.
48. Duncan A, Meek JH, Clemence M, et al. Measurement of cranial optical path length as a function of age using phase resolved near infrared spectroscopy. *Pediatr Res*. 1996;39(5):889–94.
49. Sappia MS, Hakimi N, Horschig JM. Signal quality index: an algorithm for quantitative assessment of functional near infrared spectroscopy signal quality. *Biomed Opt Express*. 2020;11(11):6733.
50. Radel R, Brisswalter J, Perrey S. Saving mental effort to maintain physical effort: a shift of activity within the prefrontal cortex in anticipation of prolonged exercise. *Cogn Affect Behav Neurosci*. 2017;17(2):305–14.
51. Scholkmann F, Holper L, Wolf U, Wolf M. A new methodical approach in neuroscience: assessing inter-personal brain coupling using functional near-infrared imaging (fNIRI) hyperscanning. *Front Hum Neurosci*. 2013;7:813.
52. Pinti P, Tachtsidis I, Hamilton A, et al. The present and future use of functional near-infrared spectroscopy (fNIRS) for cognitive neuroscience. *Ann N Y Acad Sci*. 2020;1464(1):5–29.
53. Fekete T, Rubin D, Carlson JM, Mujica-Parodi LR. The NIRS analysis package: noise reduction and statistical inference. *PLoS One*. 2011;6(9):e24322.
54. Feuerstein D, Parker KH, Boutelle MG. Practical methods for noise removal: applications to spikes, nonstationary quasi-periodic noise, and baseline drift. *Anal Chem*. 2009;81(12):4987–94.
55. Leys C, Ley C, Klein O, Bernard P, Licata L. Detecting outliers: do not use standard deviation around the mean, use absolute deviation around the median. *J Exp Soc Psychol*. 2013;49(4):764–6.
56. Montoye AHK, Vondrasek JD, Hancock JB. Validity and reliability of the VO₂ master pro for oxygen consumption and ventilation assessment. *Int J Exerc Sci*. 2020;13(4):1382–401.
57. Feldmann A, Schmitz RW, Erlacher D. Near-infrared spectroscopy-derived muscle oxygen saturation on a 0% to 100% scale: reliability and validity of the Moxy Monitor. *J Biomed Opt*. 2019;24(11):115001.
58. Bhamhani Y, Malik R, Mookerjee S. Cerebral oxygenation declines at exercise intensities above the respiratory compensation threshold. *Respir Physiol Neurobiol*. 2007;156(2):196–202.
59. Barrett LF, Satpute AB. Large-scale brain networks in affective and social neuroscience: towards an integrative functional architecture of the brain. *Curr Opin Neurobiol*. 2013;23(3):361–72.
60. Cheng S, Qiu X, Li S, Mo L, Xu F, Zhang D. Different roles of the left and right ventrolateral prefrontal cortex in cognitive reappraisal: an online transcranial magnetic stimulation study. *Front Hum Neurosci*. 2022;16:928077.
61. Arnsten AFT. Stress signalling pathways that impair prefrontal cortex structure and function. *Nat Rev Neurosci*. 2009;10(6):410–22.
62. Ekkekakis P, Hartman ME, Ladwig MA. When studying affective responses to exercise, the definition of “intensity” must reference homeostatic perturbations: a retort to Vollaard et al. *J Sport Exerc Psychol*. 2024;1–7.
63. Joyner MJ. Feeding the sleeping giant: muscle blood flow during whole body exercise. *J Physiol*. 2004;558(Pt 1):1.
64. Al-Rawi PG, Kirkpatrick PJ. Tissue oxygen index: thresholds for cerebral ischemia using near-infrared spectroscopy. *Stroke*. 2006;37(11):2720–5.
65. Canova D, Roatta S, Bosone D, Micieli G. Inconsistent detection of changes in cerebral blood volume by near infrared spectroscopy in standard clinical tests. *J Appl Physiol* (1985). 2011;110(6):1646–55.
66. Koga S, Poole DC, Kondo N, Oue A, Ohmae E, Barstow TJ. Effects of increased skin blood flow on muscle oxygenation/deoxygenation: comparison of time-resolved and continuous-wave near-infrared spectroscopy signals. *Eur J Appl Physiol*. 2015;115(2):335–43.
67. Hartman ME. Prefrontal NIRS signal is unaffected by forehead Doppler flux during incremental cycling exercise. *Clin Physiol Funct Imaging*. 2023;43(6):393–403.
68. Horiuchi M, Endo J, Handa-Kirihara Y. Relationship between cerebral oxygenation and skin blood flow at the frontal lobe during progressive hypoxia: impact of acute hypotension. *Adv Exp Med Biol*. 2020;1232:69–75.
69. Hayashi N, Kashima H, Ikemura T. Facial blood flow responses to dynamic exercise. *Int J Sports Med*. 2021;42(3):241–5.
70. Sato K, Ogoh S, Hirasawa A, Oue A, Sadamoto T. The distribution of blood flow in the carotid and vertebral arteries during dynamic exercise in humans. *J Physiol*. 2011;589(Pt 11):2847–56.